

SECTION 1-1

Data-Driven Society and Data Science

The University of Tokyo

Center for Research and Development on Transition to Modern Education

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Overview

In this section, we aim to achieve the following goals:

01 Significance of Data Science

Understand the core importance and impact of learning data science in our modern, data-driven society.

02 Data Utilization Process

Experience sequential data processes and master workflows (e.g., hypothesis testing, knowledge discovery, root cause analysis, and decision support).

03 The Future Society

Deepen understanding of upcoming paradigm shifts and infrastructural changes in future human society.

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Research & Development: Process Steps

Five Core Pillars of R&D

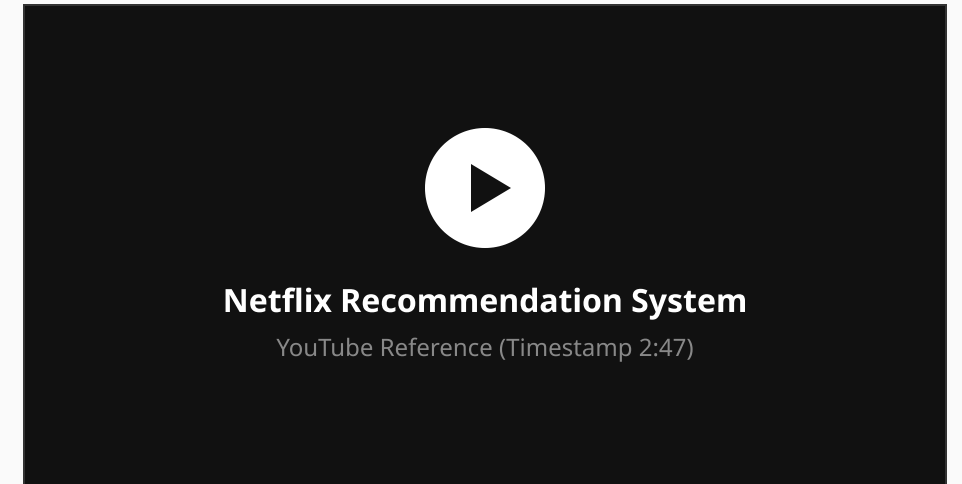
The lifecycle of data-driven research and development is strictly classified into five sequential operations: Hypothesis Verification, Knowledge Discovery, Cause Investigation, Planning, and Decision Support.

The Recommendation Ecosystem

Global streaming giants like Netflix utilize viewing history to provide hyper-personalized recommendation matrices.

Engineering Challenge

How are these systems engineered behind the scenes, and what exact data validation processes power their persistent optimization loops?



Video Link: youtu.be/0CGQvdAbNcc

Knowledge Discovery (Niche Tagging)

Beyond Broad Classifications

Media content cannot always be simplified into generic genres like "Action" or "Horror". Real audience preferences are far more nuanced.

Analyzing Viewing Behaviors

Evaluating raw user telemetry allows data systems to construct refined, granular sub-clusters that discover latent features.

Micro-Tagging at Scale

Netflix extracts hyper-specific tags like "Character-driven Suspense" or "Children's Shows featuring Vehicles", optimizing the discovery interface.



Figure: Evolutionary shift from traditional taxonomy to multi-dimensional micro-tags. (Source: YouTube youtu.be/0CGQvdAbNcc t=9:13)

R&D Planning and Decision Support

Operational Feedback Loops

Insights obtained from hypothesis testing and knowledge discovery directly feeds back into production plans and executive decision support.

Original Production Economics

The rising volume of "Original Series" across platforms (e.g., Netflix, Amazon Prime) heavily relies on historic data to model creative development.

Empirical Creative Decisions

Telemetry is used to optimize casting combinations, structural pacing, scene composition, and budget allocation, mitigating expensive creative risks.

- 1 User Behavioral Insights
- 2 Predictive Creative Modeling
- 3 Optimal Production Planning

Figure: Continuous integration of data insights into studio production decisions. (Source: YouTube youtu.be/0CGQvdAbNcc t=16:46)

New Business Models: Marketing

Data-Driven Root Cause Analysis

Modern marketing utilizes telemetry to identify the precise underlying drivers behind purchasing behaviors rather than relying on correlation.

Behavioral Log Integration

Integrating search engine footprints and query telemetry yields unparalleled predictive capabilities on consumption waves.

Spatio-Temporal GPS Analysis

High-frequency geospatial coordinates from mobile devices are actively parsed to measure demographic density and municipal mobility index during critical social emergencies.

Behavioral Log Analysis

Tracking search queries and digital footprints to model emerging trends.

Geospatial GPS Telemetry

Analyzing location patterns to trace movement dynamics and demographic density.

External References: mobaku.jp & valuesccg.com

Activity Substitution: Smart Manufacturing

Automation on the Factory Floor

Applying artificial intelligence and advanced data science on manufacturing lines represents a major shift toward high-performance activity substitution.

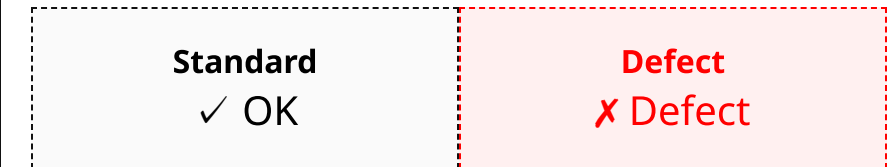
Computer Vision Quality Control

Major food manufacturers (e.g., Kewpie) utilize AI computer vision systems directly within raw ingredient sorting processes to analyze quality in real time.

Automated Sorting

Deep learning models identify defects, anomalies, and foreign objects in raw materials (like potatoes), mimicking and replacing human visual sorting with 24/7 consistency.

AI Inspection Telemetry



Sorting Threshold: 98.4% Confidence

Figure: Simulated conveyor-line computer vision defect detection dashboard. [BrainPad Case Study](<https://ai.brainpad.co.jp/case-study/194/>)

Activity Substitution: Inventory Management

Smart Stock Control

Analyzing real-time computer vision feeds in warehouse and store shelf environments completely automates standard physical inventory audits.

Demographic Profiling via Security Feeds

Security camera streams can be cross-analyzed to evaluate customer metrics such as age and gender demographics while strictly maintaining baseline data protection standards.

Synergistic Demand Forecasting

Combining shelf inventory levels with demographic purchasing patterns yields powerful, context-aware demand forecasting models that drastically optimize supply chains.

System Synergy

Camera Telemetry	Real-Time Shelf Stocks
Demographics	Customer Profiling
Integrated Output	Predictive Demand Planning

Figure: Synergistic integration of visual monitoring and behavioral metrics. [Case Study: Arts & Crafts](<https://www.arts-crafts.co.jp/post-1997/>)

Activity Substitution: Unmanned Retail

Technological Convergence

Synergizing automated self-checkout technology with advanced, camera-based inventory and tracking processes enables completely unmanned retail operations.

Addressing Demographic Challenges

This paradigm represents a critical infrastructural solution for countries experiencing rapid demographic aging, severe labor shortages, and rural depopulation.

Pioneering Deployments

Early commercial deployments include automated pilot stores (e.g., Lawson, Monotaro AI stores), proving the viability of labor-minimized commerce models.

Labor Optimization

Combining checkout automation and image tracking yields minimum staffing requirements.

*Figure: Unmanned store deployments and technological synergy.
[Techfirm Retail Blog](<https://www.techfirm.co.jp/blog/unmanned-stores-in-japan>)*

AI Implementation: Autonomous Vehicles

Integration of Multi-AI Technologies

Autonomous driving technology serves as an easily understandable, comprehensive model for the societal implementation of modern AI.

Replacing Human Environmental Perception

To control a vehicle safely, simple execution of driving functions is not enough. The vehicle must accurately interpret video feeds from external camera inputs to perform advanced situations analysis.

Target Objects of Computer Vision

The core of safety hinges on the rapid, real-time recognition of obstacles, pedestrian movement, and international road traffic regulations.

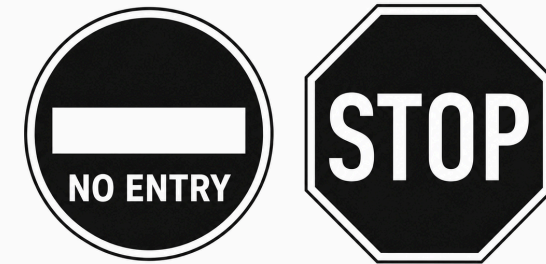


Figure: Simulated recognition targets for computer vision models (e.g., STOP and NO ENTRY regulations).

Connected Cars: Shared Network Telemetry

Real-Time Context Adjustments

To perform accurate situation analysis, autonomous vehicles must continuously parse changing environments (e.g., localized weather, accidents, and fuel/charging stations).

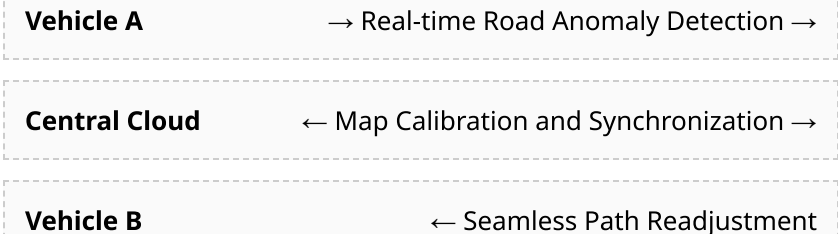
Collective Telemetry Over Isolation

Rather than operating in isolated computation loops, sharing telemetry across multiple vehicles creates far superior, crowd-sourced map precision.

The Connected Car Architecture

This strategic vision of inter-vehicular communication and centralized server telemetry aggregation is termed the "Connected Car" paradigm.

Inter-Vehicle Shared Network



*Figure: Connected Car data-exchange architecture. [Ministry of Internal Affairs and Communications]
(https://www.soumu.go.jp/main_content/000501374.pdf)*

Connected Cars: NaviCon System

Real-Time Destination Recommendation

Connected vehicles open rich possibilities for localized, context-aware point-of-interest (POI) and destination recommendations while driving.

Seamless Smartphone Integration

Interfacing smartphone navigation directly with vehicle hardware allows dynamic route adjustments and on-the-fly spot recommendations based on live travel telemetry.

Industry Implementation: DENSO NaviCon

Developed by DENSO, the NaviCon application serves as a prime industry model, enabling seamless synchronization of smartphone-selected targets with vehicle infotainment units.

NaviCon System Architecture

Smartphone App (POI
selection)



Sync

Car Infotainment
(Navigation)

One-touch target transmission protocols

*Figure: Dynamic POI transmission architecture. [DENSO NaviCon]
(<https://navicon.com/user/support/howto/index.html>)*

The Data-Driven Society Concept

Harnessing Big Data

As demonstrated by autonomous and connected vehicles, aggregating massive datasets via ubiquitous IoT devices serves as the primary engine for providing advanced AI-driven public services.

Definition of Data-Driven Society

A "Data-Driven Society" refers to a societal ecosystem where big data collection and algorithmic decision-making actively dictate resource allocation and social optimization.

Foundational Vision for the Future

This concept acts as the cornerstone for next-generation economic, industrial, and administrative strategies globally.

Socio-Technical Data Cycle

Physical Space	→ IoT Telemetry Generation →
Cyber Space	← Algorithmic Analysis (AI) →
Value Creation	← Societal Optimization & Feedback

Figure: Systematic flow of continuous data leveraging. [Ministry of Economy, Trade and Industry (METI)]
(https://www.meti.go.jp/committee/sankoushin/shojo/johokeizai/report_001.html)

Science & Technology Basic Plan

The Five-Year Legislative Cycle

In Japan, the Science and Technology Basic Law mandates the formulation of the Science and Technology Basic Plan every five years to anchor national research policies.

Coherent Historical Phases

These strategic phases systematically dictate funding parameters and cross-institutional priorities across public, private, and academic sectors.

The Birth of Society 5.0

First proposed as the ideal blueprint for Japan's future under the 5th Basic Plan (FY2016-FY2020), this vision sets the definitive tone for national innovation.

The 5th Basic Plan

Fiscal Years 2016 – 2020

Introduction and Definition of Society 5.0

The 6th Basic Plan

Fiscal Years 2021 – 2025

Focus on rapid social implementation & innovation

*Source: [Cabinet Office, Government of Japan]
(https://www8.cao.go.jp/cstp/society5_0/)*

SOCIETY 1.0 – 3.0: Historical Transitions

Society 1.0: Hunting & Gathering

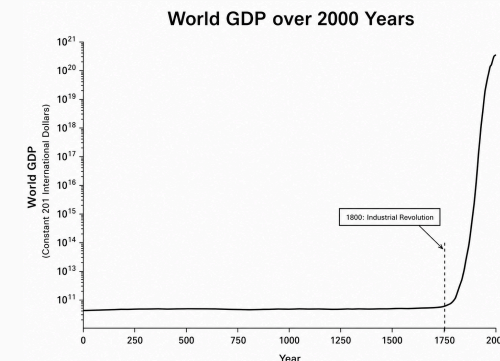
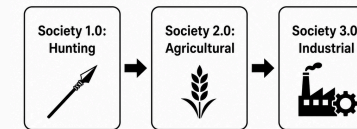
The primordial stage of human civilization, sustained entirely by direct collection of wild plants and cooperative hunting in nature.

Society 2.0: Agricultural Settlement

Triggered by the agricultural settlement revolution. Established permanent living and agricultural cultivation, dominant up to the late 18th century.

Society 3.0: Industrial Revolution

Initiated by the Industrial Revolution in the late 18th century. Shifted humanity toward factory mass production and fossil fuel energy utilization (18th to late 20th century).



SOCIETY 4.0: The Information Society

SOCIETY 4.0 Definition

The "Information Society" (representing modern society) succeeded Industrial Society, defined fundamentally by computing and data networks.

The Proliferation of Web & Internet

The rapid adoption of computers and internet connectivity catalyzed a hyper-connected public landscape, yielding massive data growth.

The "Data Overflow" Trap

In this era, humanity has essentially become overwhelmed by big data, with information retrieval and manual synthesis posing heavy cognitive and structural burdens.

Cognitive Overload

As the volume of unstructured data expands exponentially, human manual search and synthesis limits efficiency.

*Source Material: [Cabinet Office, Government of Japan]
(https://www8.cao.go.jp/cstp/society5_0/)*

SOCIETY 5.0: The Cyber-Physical Concept

A New Structural Interface

Unlike Society 4.0, which relies on humans manually searching, analyzing, and acting upon cloud-sourced knowledge, Society 5.0 automates this loop.

Deep Cyber-Physical Fusion

In this paradigm, extensive IoT sensor arrays map physical-space dynamics into high-performance cyber-space computing environments instantly.

AI-Integrated Execution

Advanced artificial intelligence models parse this cyber representation to deliver automated optimizations directly back into physical reality, breaking previous structural limits.

SOCIETY 4.0 Interface

Cyber Information Space → Manual Human Search → Physical World
Action

SOCIETY 5.0 Integration

Physical Space (Sensors) → Cyber Space (Big Data & AI) → Automated
Physical Navigation

Figure: Evolution of Cyber-Physical workflow orchestration. [Cabinet Office](https://www8.cao.go.jp/cstp/society5_0/)

SOCIETY 5.0: The Four Strategic Pillars

01

Omni-Connected IoT Synergy

Bridging isolated information spaces via unified IoT frameworks. Promotes massive data synergy and creates entirely new value streams.

02

Demographic Innovation

Leveraging systematic technology and robotic automation to address the socioeconomic challenges of aging populations and regional depopulation.

03

Cognitive Offloading via AI

Shifting the burden of data searching and curation onto autonomous AI models, providing contextual intelligence exactly when needed.

04

Physical Capability Expansion

Drastically expanding human capabilities, transport logistics, and mechanical safety through autonomous driving fleets and assistive robotics.

Implementation Platform: Smart Cities

Precursor to Society 5.0

Smart Cities serve as the pioneering sandbox, testbed environments, and baseline infrastructure for realizing the strategic objectives of Society 5.0.

Public-Private Partnerships (PPP)

In August 2018, the Government of Japan established the Smart City Public-Private Partnership Platform to coordinate administrative legislation and tech ecosystems.

Unified Technological Integration

These initiatives sync municipal transit, public administration, healthcare, and energy sectors into unified IoT structures.

Partnership Platform

Connecting public administration, academic labs, and tech giants to construct collaborative, data-driven municipal environments.

*Platform Reference: [Smart City PPP Portal]
(<https://www.mlit.go.jp/scpf/>)*

National Strategy: The Super City Law

Accelerating Societal Transition

To scale up smart cities into highly integrated "Super Cities" (Society 5.0), Japan passed landmark legislation to execute cross-cutting regulatory reforms.

Amendment Passage (May 2020)

The Diet passed the Bill to Partially Amend the National Strategic Special Zones Act (commonly referred to as the Super City Law), enabling localized regulatory bypasses.

Synergistic Multi-Domain Services

Rather than optimizing single verticals, the law coordinates data links across multiple life-supporting services (finance, medical, energy) to establish clear demographic value evaluated by local citizens.

The Super City Law

Enabling comprehensive, cross-sector regulatory deregulation to accelerate high-dimensional IoT coordination on national scale.

Source: [National Strategic Special Zones Open Lab]

(<https://www.kantei.go.jp/jp/singi/tiiki/kokusentoc/supercity/openlabo/supercitycontents.html>)

Smart City Model: Toyota Woven City

Human-Centered Living Laboratory

Toyota's Woven City represents a prime, privately-funded real-world testbed designed specifically to test emerging technologies in a real residential setting.

Three Core Philosophy Pillars

The project centers around three immutable core values: Human-Centered Design, A Living Laboratory for experimental technologies, and An Ever-Evolving, Unfinished City.

Demographic Scaling

Starting with 360 residents and expanding to over 2,000 (including researchers and engineers), the city constructs a live data framework to solve critical urban challenges.

Toyota Woven City

An experimental smart city serving as a real-world sandbox for automated mobility, personal robotics, smart homes, and advanced AI systems.

Project Hub: [Woven City Global Portal](<https://www.woven-city.global/jpn>)

The 6th Basic Plan (Innovation Strategy)

A Brand-New Nomenclature

Covering FY2021-FY2025, Japan renamed its legislative policy to the "Science, Technology, and Innovation Basic Plan" to emphasize social implementation.

Rapid Social Deployment

The core objective shifts from basic laboratory research toward rapid, crisis-aware social implementation loops to realize Society 5.0 immediately.

Broad Structural Goals

Prioritizes robust R&D investment parameters, global talent cultivation, collaborative socio-technical dialogue, and resilient systems capable of tackling public health and national safety crises.

Concrete Materialization of Society 5.0

Crisis-Aware Speed in Social Implementation

Cultivation of Global Research Talent

Enhanced Public-Private R&D Investment

*Source: [CSTP Innovation Basic Plan]
(<https://www8.cao.go.jp/cstp/kihonkeikaku/index6.html>)*